

The term basal ganglia is used to denote a number of subcortical nuclear masses that lie in the inferior part of the cerebral hemisphere, in close relationship with the internal capsule (Fig. 24.1; see Fig. 25.43). The traditional definition of the basal ganglia included the corpus striatum, claustrum and amygdaloid complex. The term has now been restricted to the corpus striatum and, according to some authorities, other nuclei in the diencephalon and midbrain (principally the subthalamic nucleus, substantia nigra, pedunculopontine nucleus) that collectively form a functional complex involved in the control of movement and motivational aspects of behaviour (Jankovic 2012). The function of the claustrum is unknown; the amygdala is more closely related to the limbic system and is, therefore, described in that context.

Disorders of the basal ganglia are principally characterized by abnormalities of movement, muscle tone and posture. There is a wide spectrum of clinical presentations ranging from poverty of movement and hypertonia at one extreme (typified by Parkinson's disease) to abnormal involuntary movements (dyskinesias) at the other. The underlying pathophysiological mechanisms that mediate these disorders have been much studied in recent years and are better understood than for any other type of complex neurological dysfunction (Penney and Young 1986, Jankovic 2012). This has led to the introduction of new rational

therapeutic strategies for both medical and neurosurgical treatments of movement disorders.

CORPUS STRIATUM

The corpus striatum consists of the caudate nucleus, putamen and globus pallidus (Fig. 24.2). Because of their close proximity, the putamen and globus pallidus were once considered as an entity, the lentiform (lenticular) complex or nucleus. However, although the name has been retained in gross anatomical terminology and in some compound names (e.g. sublentiform, retrolenticular), the putamen and globus pallidus have quite different connections. Rather, it is now known that the putamen and caudate nucleus share a common chemoarchitecture and connections, and they are referred to jointly as the neostriatum, or simply the striatum.

The striatum is considered to be the principal 'input' structure of the basal ganglia since it receives the majority of afferents from other parts of the neuraxis. Its principal efferent connections are to the globus pallidus and pars reticularis of the substantia nigra. The globus pallidus and, more particularly, its medial segment, together with the pars reticularis of the substantia nigra, is regarded as the main 'output' structure because it is the source of basal ganglia efferent fibre projections, mostly directed to the thalamus.

LENTIFORM COMPLEX

The lentiform complex (see Figs 16.7, 24.1; Fig. 24.3) lies deep to the insular cortex, with which it is roughly coextensive, although they are separated by a thin layer of white matter and by the claustrum. The latter splits the insular subcortical white matter to create the extreme

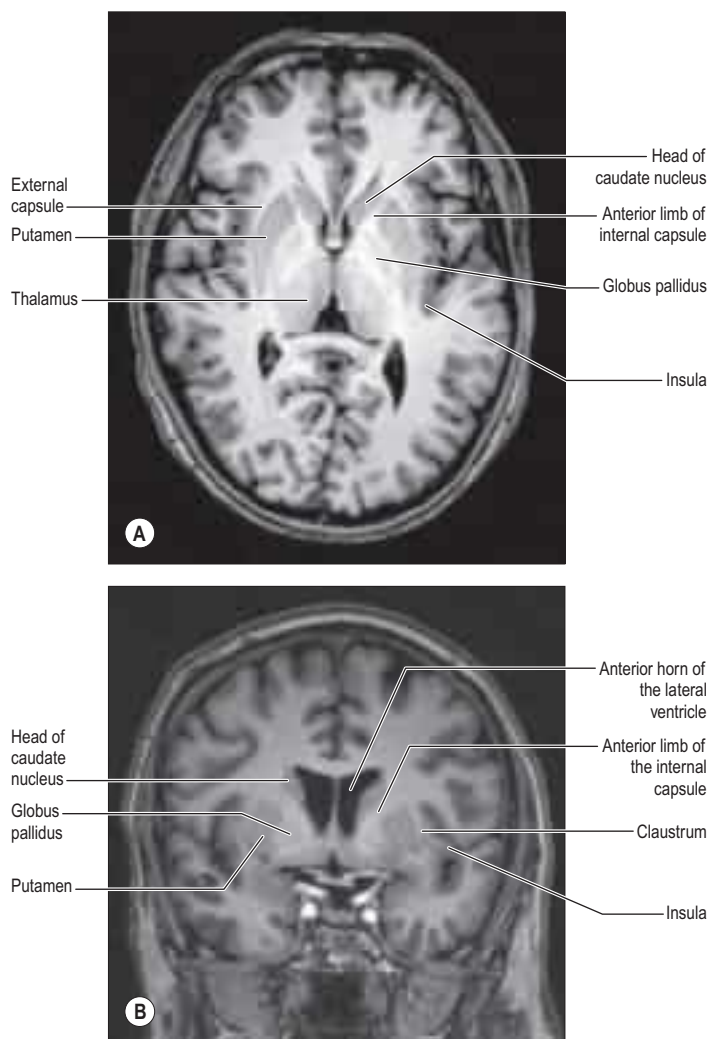


Fig. 24.1 Axial (A) and coronal (B) magnetic resonance images of the brain showing the basal ganglia, thalamus and internal capsule. (Courtesy of Alan Jackson.)

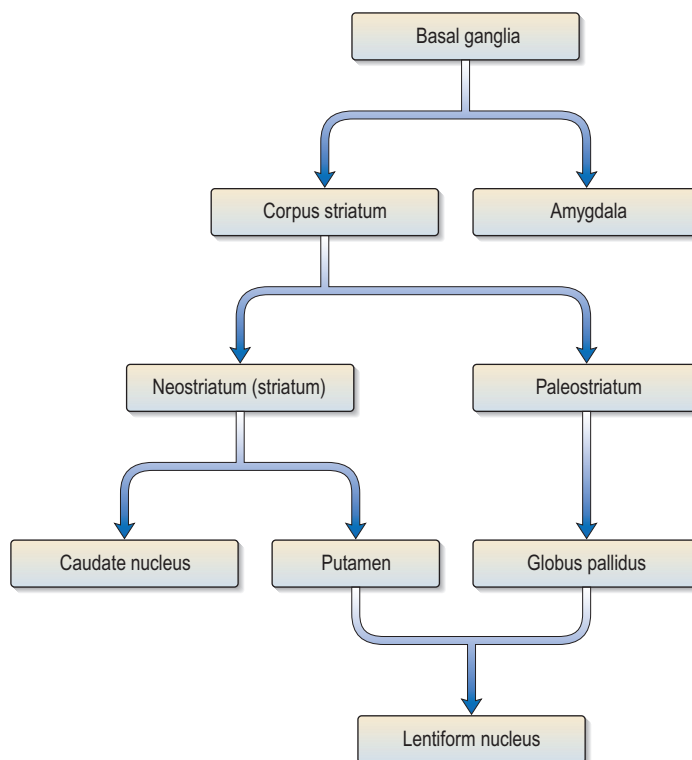


Fig. 24.2 Terminology and relationships of structures forming the basal ganglia.

and external capsules; the external capsule separates the claustrum from the putamen. The internal capsule separates the lentiform complex from the caudate nucleus.

The lentiform complex consists of the laterally placed putamen and the more medially placed globus pallidus (pallidum), which are separated by a thin layer of fibres, the external medullary lamina. The globus pallidus is itself divided into two segments, a lateral (external) segment and a medial (internal) segment, separated by an internal medullary lamina. The two segments have distinct afferent and efferent connections. Inferiorly, a little behind the fundus striati, the lentiform complex is grooved by the anterior commissure, which connects inferior parts of the temporal lobes and the anterior olfactory cortex of the two sides (see Fig. 24.6). The area above the commissure is referred to as the dorsal pallidum, and that below it as the ventral pallidum.

STRIATUM

The striatum consists of the caudate nucleus, putamen and ventral striatum, which are all highly cellular and well vascularized. The vast bulk of the caudate nucleus and putamen are often referred to as the

dorsal striatum. A smaller inferomedial part of the rostral striatum is referred to as the ventral striatum, and includes the nucleus accumbens. The caudate and putamen are traversed by numerous small bundles of thinly myelinated, or unmyelinated, small-diameter axons, which are mostly striatal afferents and efferents. They radiate through the striatal tissue as though converging on, or radiating from, the globus pallidus. The bundles are occasionally referred to by the archaic term 'Wilson's pencils' and they account for the striated appearance of the corpus striatum.

CAUDATE NUCLEUS

The caudate nucleus is a curved, tadpole-shaped mass. It has a large anterior head, which tapers to a body, and a down-curving tail (Fig. 24.4). The head is covered with ependyma and lies in the floor and lateral wall of the anterior horn of the lateral ventricle, in front of the interventricular foramen. The tapering body is in the floor of the body of the ventricle, and the narrow tail follows the curve of the inferior horn, and so lies in the ventricular roof, in the temporal lobe. Medially, the greater part of the caudate nucleus abuts the thalamus, along a junction that is marked by a groove, the sulcus terminalis. The sulcus contains the stria terminalis, lying deep to the ependyma (Fig. 24.5).

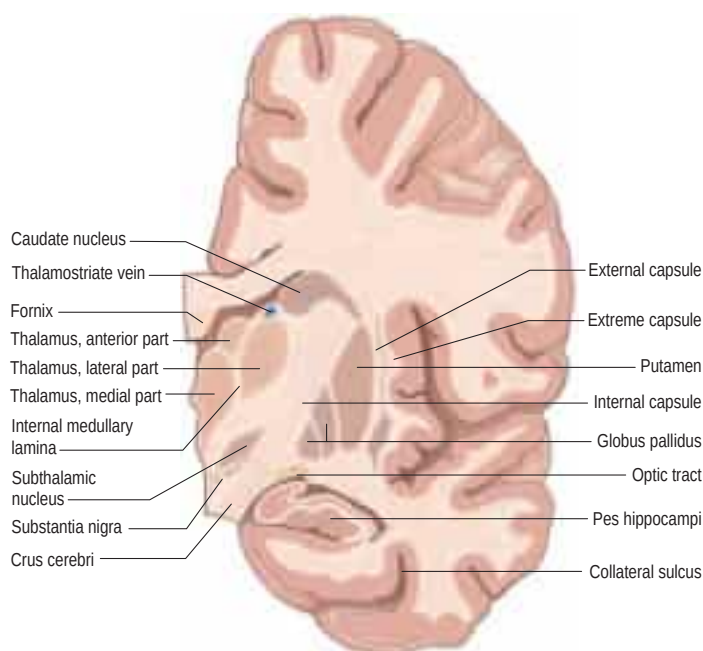


Fig. 24.3 The anterior aspect of a coronal section through the left cerebral hemisphere.

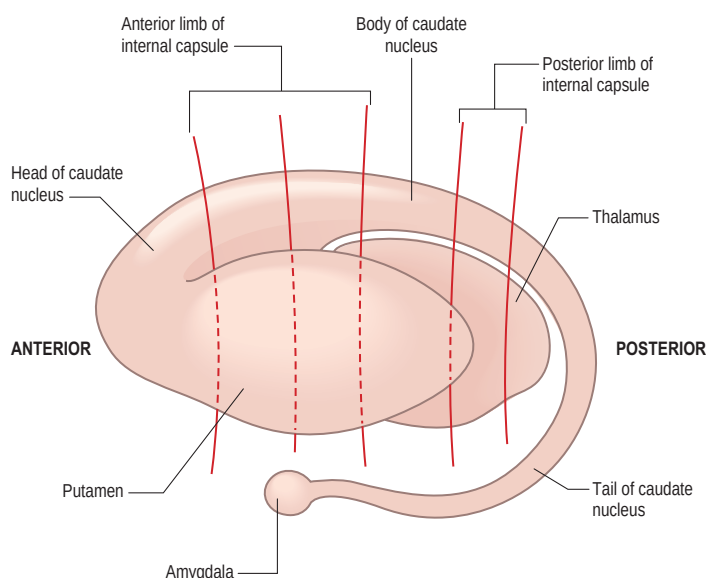


Fig. 24.4 The striatum within the left cerebral hemisphere. (With permission from Crossman AR, Neary D 2010 Neuroanatomy, 4th edn. Edinburgh: Churchill Livingstone.)

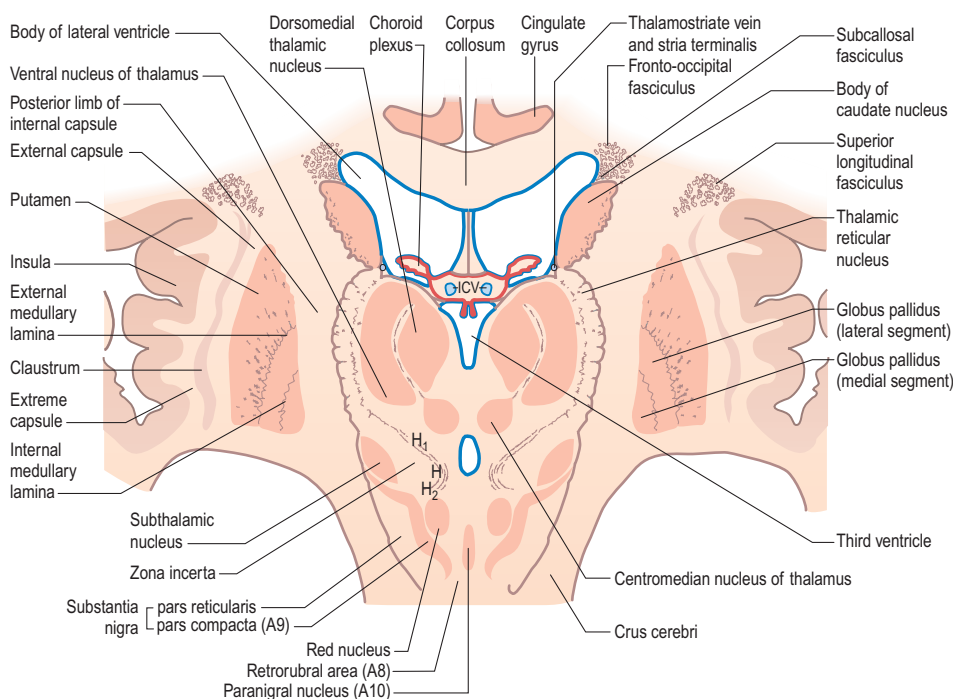


Fig. 24.5 An oblique section through the diencephalon and basal ganglia. Abbreviations: ICV, internal cerebral veins; H, H₁, H₂, subthalamic fields of Forel.

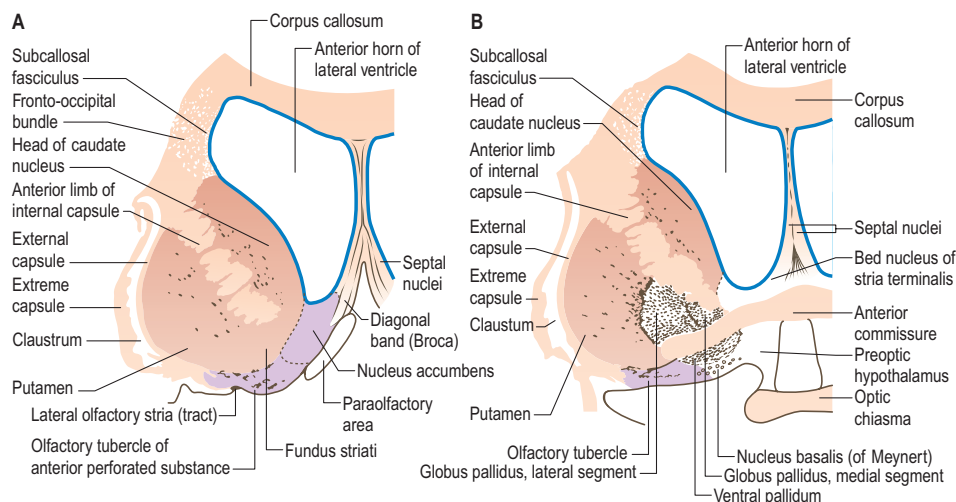


Fig. 24.6 Coronal sections through the corpus striatum and anterior perforated substance. **A** is anterior to **B**.

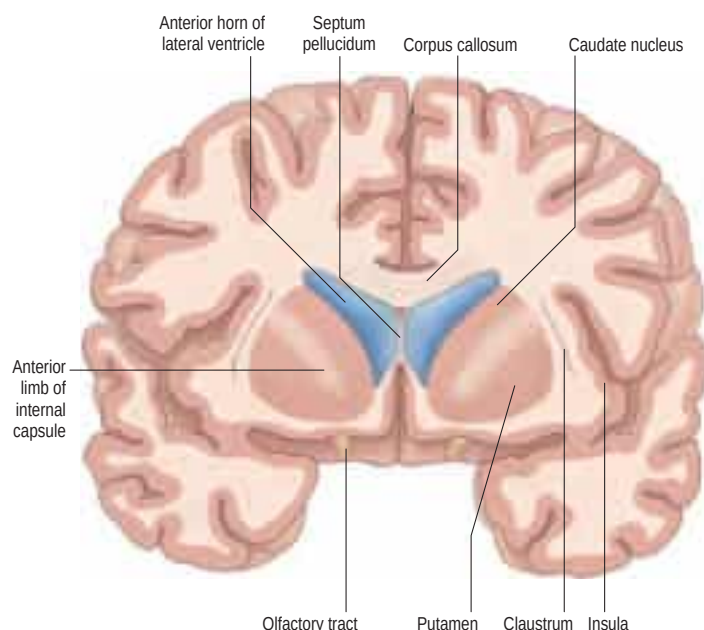


Fig. 24.7 The posterior aspect of a coronal section through the anterior horn of the lateral ventricles.

The sulcus terminalis is especially prominent anterosuperiorly (because of the large size of the head and body of the caudate nucleus relative to the tail), and here the stria terminalis is accompanied by the thalamostriate vein.

The corpus callosum lies above the head and body of the caudate nucleus. The two are separated laterally by the fronto-occipital fasciculus, and medially by the subcallosal fasciculus, which caps the nucleus (see Fig. 24.5; Fig. 24.6). The caudate nucleus is largely separated from the lentiform complex by the anterior limb of the internal capsule (see Figs 24.1, 24.6; Fig. 24.7). However, the inferior part of the head of the caudate becomes continuous with the most inferior part of the putamen immediately above the anterior perforated substance; this junctional region is sometimes known as the fundus striati (see Fig. 24.6). Variable bridges of cells connect the putamen to the caudate nucleus for most of its length. They are most prominent anteriorly, in the region of the fundus striati and the head and body of the caudate nucleus, where they penetrate the anterior limb of the internal capsule (see Fig. 24.7). In the temporal lobe, the anterior part of the tail of the caudate nucleus becomes continuous with the posteroinferior part of the putamen.

VENTRAL STRIATUM

The ventral striatum consists of the nucleus accumbens and the olfactory tubercle. In front of the anterior commissure, much of the grey matter of the anterior perforated substance, and especially the olfactory tubercle, is indistinguishable from, and continuous with, the fundus striati, in terms of cellular composition, histochemistry and interconnections. The caudate nucleus is continuous medially with the

nucleus accumbens (see Fig. 24.6), which abuts the nuclei of the septum, close by the paraolfactory area, diagonal band of Broca and the fornix.

The nucleus accumbens receives a dopaminergic innervation from the midbrain ventral tegmental area (cell group A10). It is believed to represent the neural substrate for the rewarding effects of several classes of drugs of abuse and is, therefore, a major determinant of their addictive potential. The experimental observation that the locomotor activating effects of psychomotor stimulant drugs such as amphetamine and cocaine (which act presynaptically on dopaminergic neurones to enhance dopamine release or block its reuptake, respectively) are dependent on dopamine transmission in the nucleus accumbens, led to the hypothesis that the reinforcing or rewarding properties of these drugs are mediated by the mesolimbic dopamine system.

STRIATAL CONNECTIVITY

Neurones of both dorsal and ventral striatum are mainly medium-sized multipolar cells. They have round, triangular or fusiform somata, mixed with a smaller number of large multipolar cells. The ratio of medium to large cells is at least 20:1. The large neurones have extensive spherical or ovoid dendritic trees up to 60 μ m across. The medium-sized neurones also have spherical dendritic trees, approximately 20 μ m across, which receive the synaptic terminals of many striatal afferents. The dendrites of both medium and large striatal cells may be either spiny or non-spiny. The most common neurone (approximately 75% of the total) is a medium-sized cell with spiny dendrites (so-called medium spiny neurones). These cells utilize γ -aminobutyric acid (GABA) as their transmitter and also express the genes coding for either enkephalin or substance P/dynorphin, depending on their efferent connections. Enkephalinergic neurones express D2 dopamine receptors. Substance P/dynorphin neurones express D1 receptors. These neurones are the major, and perhaps exclusive, source of striatal efferents to the pallidum and substantia nigra pars reticularis. The other medium-sized striatal neurones are aspiny, and are intrinsic cells that contain acetylcholinesterase (AChE), choline acetyltransferase (CAT) and somatostatin. Large neurones with spiny dendrites contain AChE and CAT; most, perhaps all, are intrinsic neurones. Aspiny large neurones are all intrinsic neurones.

Intrinsic synapses are probably largely asymmetric (type II), while those derived from external sources are symmetric (type I). The aminergic afferents from the substantia nigra, raphe and locus coeruleus all end as profusely branching axons with varicosities, which contain dense-core vesicles (the presumed store of amine transmitters). Many of these varicosities have no conventional synaptic membrane specializations, and may release transmitter in a way analogous to that found in peripheral postsynaptic sympathetic axons.

Neuroactive chemicals, whether intrinsic or derived from afferents, are not distributed uniformly in the striatum. For example, 5-HT (5-hydroxytryptamine, serotonin) and glutamic acid decarboxylase (GAD) concentrations are highest caudally, while substance P, acetylcholine (ACh) and dopamine are highest rostrally. However, there is a finer-grain neurochemical organization showing that the striatum consists of a mosaic of islands or striosomes (sometimes referred to as patches), each 0.5–1.5 mm across, packed into a background matrix. Striosomes contain substance P and enkephalin. During development,

the first dopamine terminals from the substantia nigra are found in striosomes. Although this exclusivity does not persist after birth, striosomes in the adult caudate nucleus still contain a higher concentration of dopamine than the matrix. The latter contains ACh and somatostatin, and is the target of thalamostriatal axons. Receptors for at least some neurotransmitters are also differentially distributed. For example, opiate receptors are found almost exclusively within striosomes, and muscarinic receptors predominantly so. Moreover, the distribution of neuroactive substances within the striosomes is not uniform. In humans, the striosome/matrix patchwork is more evident in the caudate than the putamen, the latter consisting predominantly of matrix.

All afferents to the striatum terminate in a mosaic manner. The size of a cluster of terminals is usually 100–200 μm across. Some afferent terminal clusters are not arranged in register with the clear striosome/matrix distributions seen in nigrostriatal and thalamostriatal axons. In general, afferents from the neocortex end in striatal matrix and those from the allocortex end in striosomes. However, the distinction is not absolute: although afferents from the neocortex arise in layers V and VI, those from the superficial part of layer V end predominantly in striatal matrix, whereas those from deeper neocortex project to striosomes. Striatal cell bodies that are the sources of efferents also form clusters but, again, are not uniformly related to striosomes. For example, the cell bodies of some striatopallidal and striatonigral axons lie clustered within striosomes, but others lie outside them, still in clusters. The neurones and neuropil of the ventral striatum are essentially similar to those of the dorsal striatum, but the striosomal/matrix organization is less well defined and seems to consist predominantly of striosomes.

The major connections of the striatum are summarized in [Figure 24.8](#). Although the connections of the dorsal and ventral divisions overlap, the generalization can be made that the dorsal striatum is predominantly connected with motor and associative areas of the cerebral cortex, whilst the ventral striatum is connected with the limbic system and orbitofrontal and temporal cortices. For both dorsal and ventral striatum, the pallidum and substantia nigra pars reticularis are key efferent structures. The fundamental arrangement is the same for both divisions. The cerebral cortex projects to the striatum, which in turn projects to the pallidum and substantia nigra pars reticularis. Efferents from the pallidum and substantia nigra pars reticularis influence the cerebral cortex (either the supplementary motor area or prefrontal and cingulate cortices via the thalamus) and the superior colliculus (see below).

The entire neocortex sends glutamatergic axons to the ipsilateral striatum. Previously, these axons were thought to be collaterals of cortical efferents to other regions, but it is now known that they arise exclusively from small pyramidal cells in layers V and VI. It has also been

suggested that some of the cells of origin lie in the supragranular 'cortical association' layers II and III. The projection is organized topographically. The greater part of the input from the cerebral cortex to the dorsal striatum is derived from the frontal and parietal lobes, with that from the occipitotemporal cortex being relatively small. Thus, the orbitofrontal association cortex projects to the inferior part of the head of the caudate nucleus, which lies next to the ventral striatum. The dorsolateral frontal association cortex and frontal eye fields project to the rest of the head of the caudate nucleus, and much of the parietal lobe projects to the body of the nucleus. The somatosensory and motor cortices project predominantly to the putamen. Their terminals establish a somatotopic pattern, in which the lower body is represented laterally and the upper body is represented medially. The motor cortex is unique in also sending axons through the corpus callosum to the opposite putamen, where they end with the same spatial ordering. The occipital and temporal cortices project to the tail of the caudate nucleus and to the inferior putamen.

The striatum receives afferents from the polysensory intralaminar thalamus and these are more crudely organized spatially. The nucleus centralis lateralis, which receives a cerebellar input, projects to the anterior striatum (especially the caudate nucleus), while the centromedian nucleus, which receives input from both the cerebellum and internal (medial) globus pallidus, projects to the putamen.

The aminergic inputs to the caudate and putamen are derived from the substantia nigra pars compacta (dopaminergic group A9; see [Fig. 21.17](#)), the retrorubral nucleus (dopaminergic group A8), the dorsal raphe nucleus (serotonergic group B7) and the locus coeruleus (noradrenergic group A6). The nigrostriatal input is sometimes referred to as the 'mesostriatal' dopamine pathway. It reaches the striatum by traversing the H fields (of Forel) in the subthalamus and running in the medial forebrain bundle. These aminergic inputs appear to modulate the responses of the striatum to cortical and thalamic afferent influences.

Efferents from the striatum pass to both segments of the globus pallidus and to the substantia nigra pars reticularis, where they end in a topically ordered fashion. Fibres ending in either the external or internal pallidum originate from different striatal cells (see [Fig. 24.8](#)). Those projecting to the external pallidum come from neurones that co-localize GABA and enkephalin, and give rise to the so-called 'indirect pathway'. This name refers to the fact that these striatal neurones influence the activity of basal ganglia output neurones in the internal pallidum via the intermediary of the subthalamic nucleus. Other striatal neurones, which co-localize GABA and substance P/dynorphin, project directly to the internal pallidum and are, therefore, described as the 'direct pathway'.

The striatal projection to the pars reticularis of the substantia nigra also has both direct and indirect components, via the external pallidum and subthalamic nucleus ([Figs 24.9–24.10](#)). The axons of the direct striatonigral projection constitute the laterally placed 'comb' system, which is spatially quite distinct from the ascending dopaminergic nigrostriatal pathway. Striatonigral fibres end in a spatially ordered way in the pars reticularis.

The ventral striatum is the primary target of fibres from limbic cortices, including allocortex, and from limbic associated regions (see [Fig. 24.9](#)). Thus, the hippocampus (through the fornix) and orbitofrontal cortex (through the internal capsule) project to the nucleus accumbens, and the olfactory, entorhinal, anterior cingulate and temporal visual cortices project to both the nucleus accumbens and olfactory tubercle. The olfactory tubercle also receives afferents from the amygdala. The contiguity of the cortical areas that project to the ventral striatum and neighbouring dorsal striatum emphasizes the imprecise nature of the boundaries between these two divisions. All the cortical regions abut and overlap with neighbouring areas and they project to neighbouring parts of the dorsal striatum as well as to the ventral striatum. The fundus striati and ventromedial caudate nucleus about the olfactory tubercle and nucleus accumbens (see [Fig. 24.6](#)) and receive connections from the orbitofrontal cortex and, to a lesser extent, from the lateral prefrontal and anterior cingulate cortices (which also project to the contiguous head of the caudate nucleus).

This continuity of the ventral and dorsal striata, as revealed by the arrangements of corticostriatal projections, is reinforced by consideration of the aminergic inputs to the ventral striatum. They are derived from the dorsal raphe (serotonergic group B7), the locus coeruleus (noradrenergic group A6) and the ventral tegmental area (dopamine group A10), as well as the most medial part of the substantia nigra pars compacta (A9) (see [Fig. 21.17](#)). The dopamine projections constitute the so-called mesolimbic dopamine pathway, which also projects to the septal nuclei, hippocampus and amygdala, and prefrontal and cingulate cortices through the medial forebrain bundle. The lateromedial

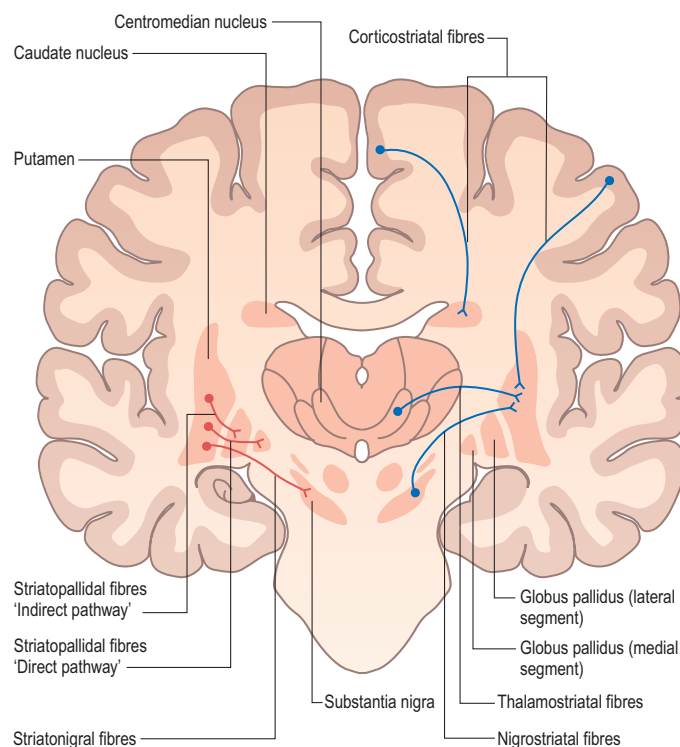


Fig. 24.8 Connections of the striatum. The major afferent projections to the striatum are shown on the right and major efferent projections from the striatum on the left.

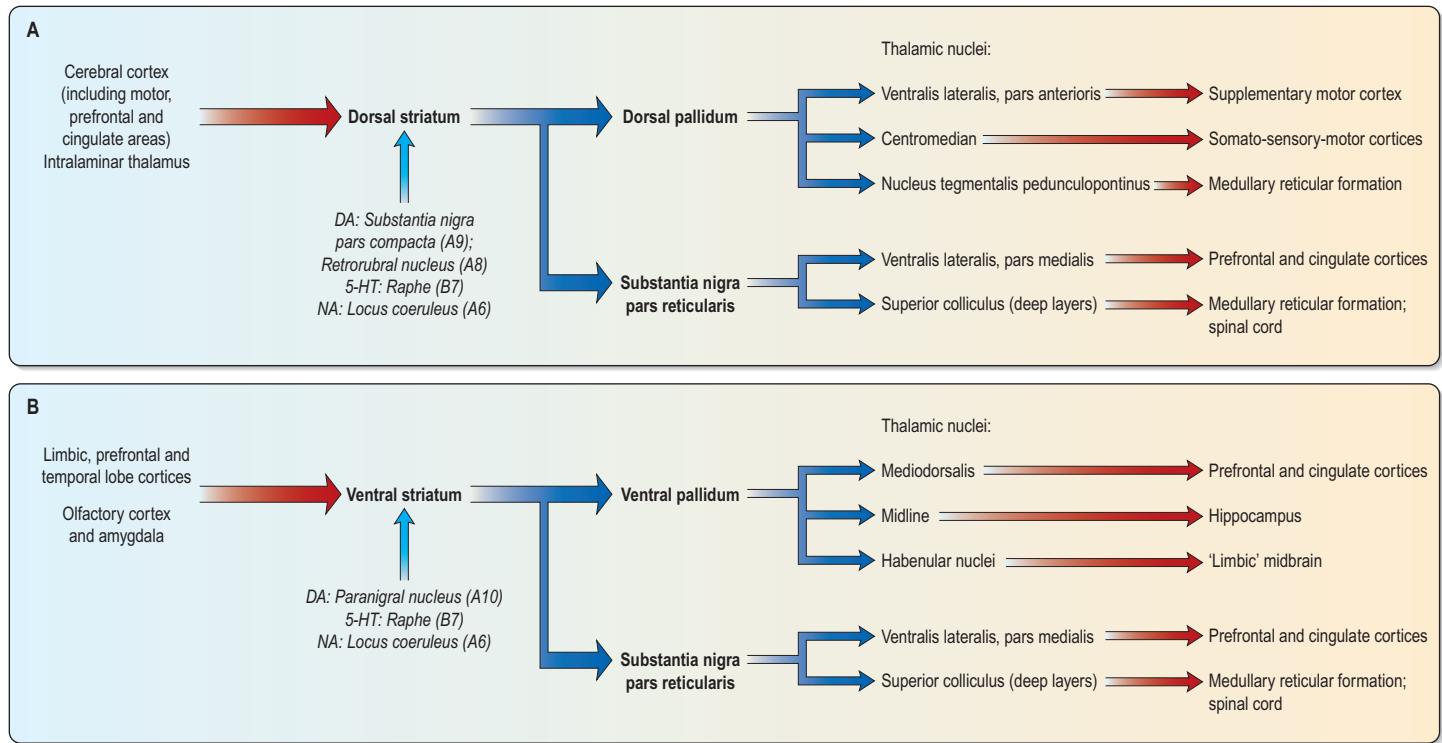


Fig. 24.9 The principal output connections of the basal ganglia derived from dorsal (A) and ventral (B) divisions of the striatum. In each case, pathways established through the pallidum are distinguished from those passing through the substantia nigra pars reticularis. Abbreviations: DA, dopamine; NA, noradrenaline (norepinephrine); 5-HT, 5-hydroxytryptamine (serotonin).

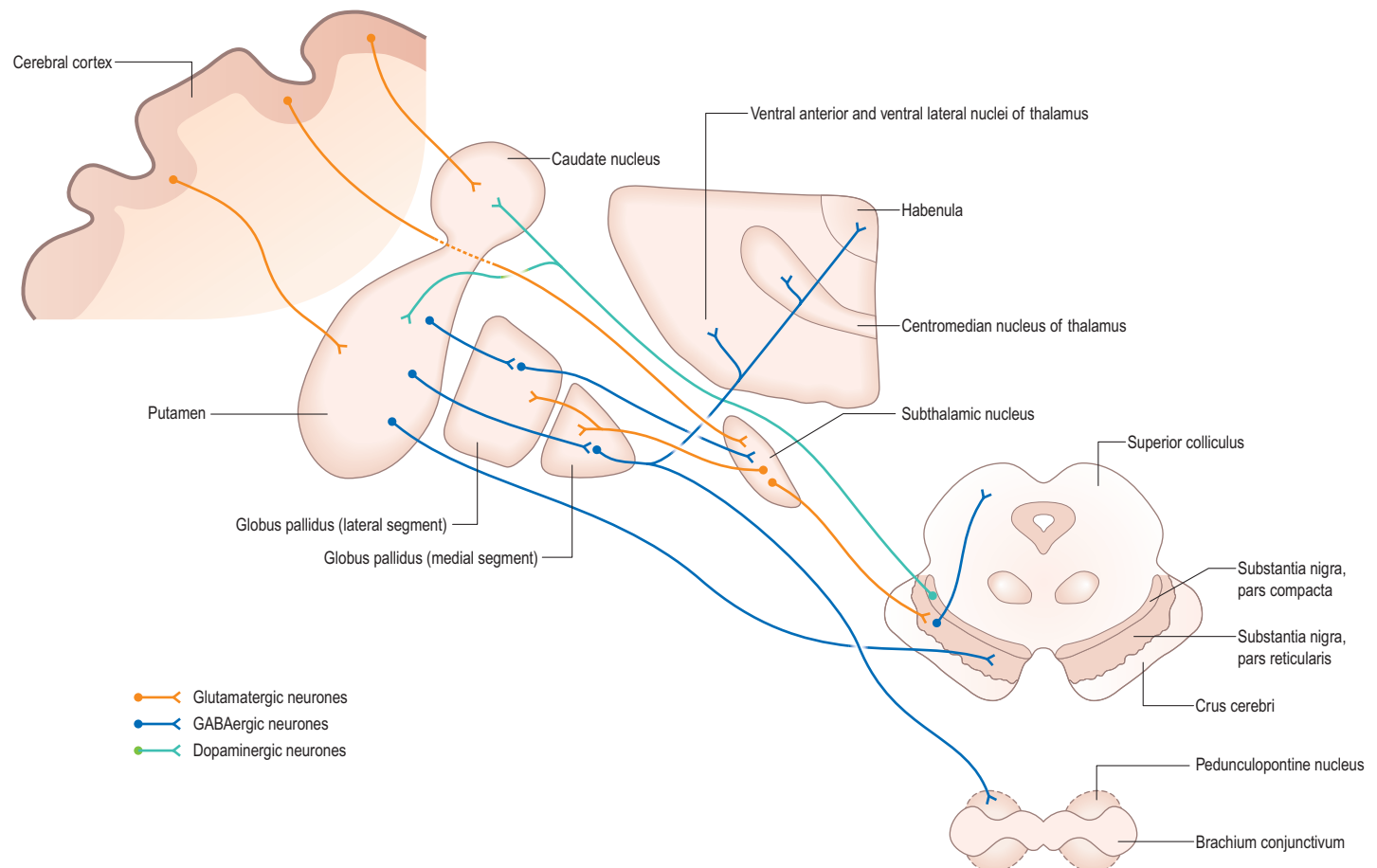


Fig. 24.10 The principal connections of the basal ganglia with the diencephalon and brainstem.

continuity of cell groups A9 and 10 is thus reflected in the relative positions of their ascending fibres in the subthalamus and hypothalamus (the H fields and medial forebrain bundle, respectively), as well as in the lateromedial topography of the dorsal and ventral striata (see Fig. 24.6), which in turn have contiguous and overlapping sources of cortical afferents.

As with the dorsal striatum, efferents from the ventral striatum project to the pallidum (in this case, the ventral pallidum) and the substantia nigra pars reticularis (see Figs 24.9–24.10). In the latter case, the connection is both direct and indirect, via the subthalamic nucleus. The projections from the pars reticularis are as described for the dorsal system, but axons from the ventral pallidum reach

the thalamic mediodorsal nucleus (which projects to cingulate and prefrontal association cortex) and midline nuclei (which project to the hippocampus). Ventral pallidal axons also reach the habenular complex of the limbic system.

The brain areas beyond the basal ganglia, substantia nigra and subthalamic nucleus to which both ventral and dorsal systems appear to project are, therefore, the prefrontal association and cingulate cortices and the deep superior colliculus.

GLOBUS PALLIDUS

The globus pallidus (pallidum) lies medial to the putamen and lateral to the internal capsule. It consists of external (lateral) and internal (medial) segments separated by an internal medullary lamina, which have substantially different connections. Both segments receive large numbers of fibres from the striatum and subthalamic nucleus. The external segment projects to the subthalamic nucleus as part of the 'indirect pathway'. The internal segment is considered to be a homologue of the pars reticularis of the substantia nigra, with which it shares similar cellular and connectional properties. Together, therefore, these structures constitute the main output of the basal ganglia to other levels of the neuraxis, principally to the thalamus and superior colliculus.

The cell density of the globus pallidus is less than one-twentieth that of the striatum. The morphology of the majority of cells is identical in the two segments. They are large multipolar GABAergic neurones that closely resemble those of the substantia nigra pars reticularis. The dendritic fields are discoid, with planes at right angles to incoming striato-pallidal axons, each of which, therefore, potentially contacts many pallidal dendrites *en passant*. This arrangement, coupled with the diameters of the dendritic fields (500 µm), suggests that a precise topographical organization is unlikely within the pallidum.

Striatopallidal fibres are of two main types. They project to either the external or the internal pallidum. Those projecting to the external segment constitute the beginning of the so-called 'indirect pathway'. They utilize GABA as their primary transmitter and also contain enkephalin. Efferent axons from neurones in the external segment pass through the internal capsule in the subthalamic fasciculus and travel to the subthalamic nucleus (Fig. 24.11).

Striatopallidal axons destined for the internal pallidum constitute the so-called 'direct pathway'. Like the indirect projection, these utilize GABA as their primary transmitter but they also contain substance P and dynorphin. Efferent axons from the internal pallidum project through the ansa lenticularis and fasciculus lenticularis (see Fig. 24.11). The former runs round the anterior border of the internal capsule and the latter penetrates the capsule directly. Having traversed the internal

capsule, both pathways unite in the subthalamic region, where they follow a horizontal hairpin trajectory, and turn upwards to enter the thalamus as the thalamic fasciculus. The trajectory circumnavigates the zona incerta and creates the so-called 'H' fields of Forel (see Figs 24.5, 24.11). Within the thalamus, pallidothalamic fibres end in the ventral anterior and ventral lateral nuclei and in the intralaminar centromedian nucleus. These in turn project excitatory (presumed glutamatergic) fibres primarily to the frontal cortex, including the primary and supplementary motor areas. The internal pallidum also projects fibres caudally to the pedunculopontine nucleus, as described below (see Fig. 24.10).

SUBTHALAMIC NUCLEUS

The subthalamic nucleus is a biconvex, lens-shaped nucleus in the subthalamus of the diencephalon. It lies medial to the internal capsule, immediately rostral to the level at which the latter becomes continuous with the crus cerebri of the midbrain (see Figs 24.5, 24.3). Within the nucleus, small interneurons intermingle with large multipolar cells with dendrites that extend for about one-tenth the diameter of the nucleus. Dorsally, the nucleus is encapsulated by axons, many of which are derived from the subthalamic fasciculus; these carry a major GABAergic projection from the external segment of the globus pallidus as part of the indirect pathway. The nucleus also receives afferents from the cerebral cortex. It is unique in the intrinsic circuitry of the basal ganglia in that its cells are glutamatergic and project excitatory axons to both the globus pallidus and substantia nigra pars reticularis. Within the pallidum, subthalamic efferent fibres end predominantly in the internal segment but many also end in the external segment.

The subthalamic nucleus plays a central role in the normal function of the basal ganglia and in the pathophysiology of basal ganglia-related disorders. Destruction of the nucleus, which occurs rarely as a result of stroke, results in the appearance of violent, uncontrolled involuntary movements of the contralateral side of the body, known as ballism (hemiballismus). The subthalamic nucleus is crucially involved in the pathophysiology of Parkinson's disease and is a target for functional neurosurgical therapy of the condition. (For a detailed description of the anatomical structure and variability of the subthalamic nucleus seen using high-field magnetic resonance imaging (MRI) with histological validation, see Massey et al (2012) and Massey and Yousry (2010).)

SUBSTANTIA NIGRA

The substantia nigra is a nuclear complex deep to the crus cerebri in each cerebral peduncle of the midbrain (see Fig. 21.16); its

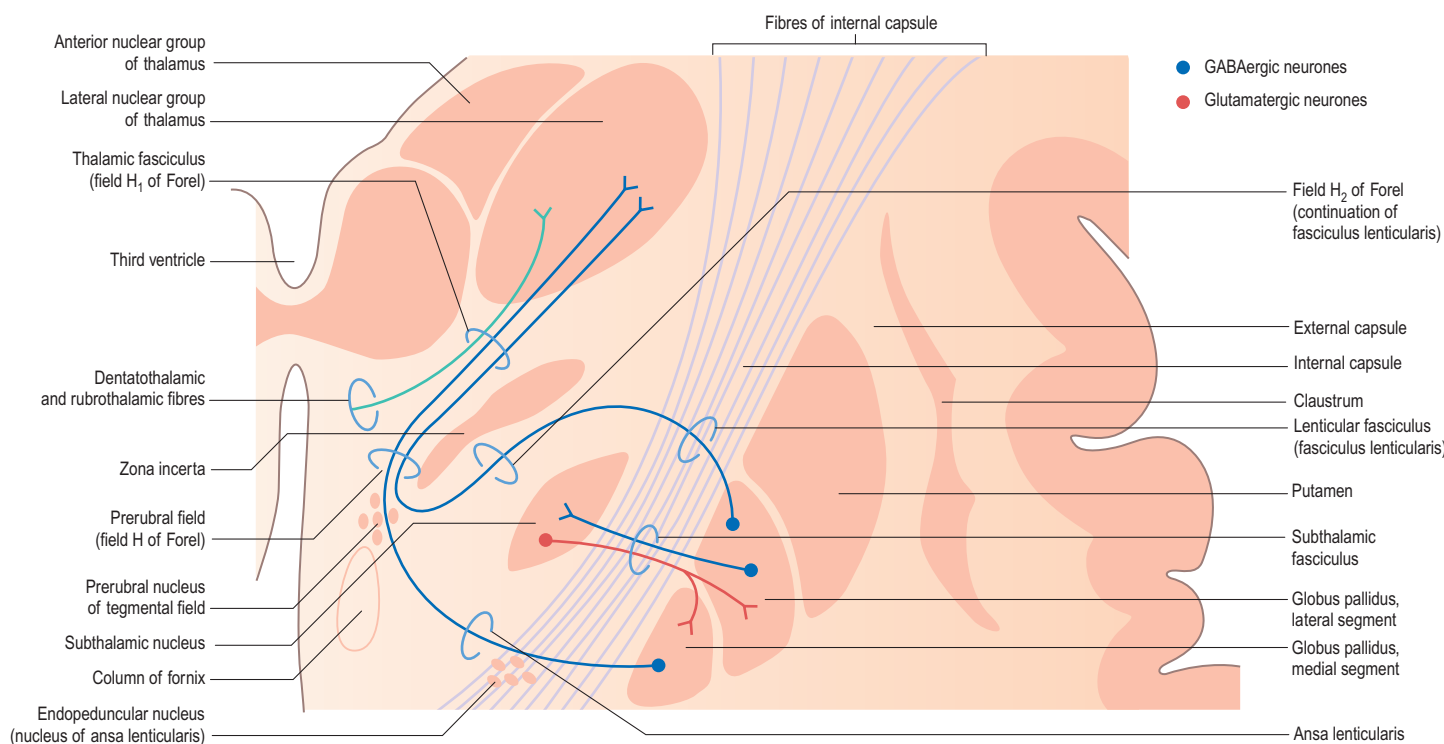


Fig. 24.11 A coronal section of the brain showing the major connections of the basal ganglia with the diencephalon.

cytoarchitecture and connections are described in [Chapter 21](#). Briefly, it consists of a pars compacta, a pars reticularis and a smaller pars lateralis. The pars compacta and pars lateralis correspond to the dopaminergic cell group A9. The pars compacta projects heavily to the caudate nucleus and putamen; lesser projections end in the globus pallidus and the subthalamic nucleus. The neurones of the pars reticularis and the internal pallidum collectively constitute the output neurones of the basal ganglia system. In Parkinson's disease, the levels of dopamine in the substantia nigra and striatum decrease dramatically as a result of the degeneration of pars compacta neurones and their terminals in the neostriatum.

PEDUNCULOPONTINE NUCLEUS

The pedunculopontine nucleus (nucleus tegmentalis pedunculopontinus) lies in the dorsolateral part of the pontomesencephalic tegmentum. Anterograde tract-tracing studies in non-human primates and rodents have revealed multiple afferent and efferent connections between the pedunculopontine nucleus and the basal ganglia, cerebellum, substantia nigra, thalamus, cerebral cortex and spinal cord.

The pedunculopontine nucleus receives GABAergic efferent fibres from the globus pallidus and substantia nigra. In animal models of Parkinson's disease, these projections are overactive and the pedunculopontine nucleus is inhibited. Taken in conjunction with other experimental evidence, this suggests that the pedunculopontine nucleus is involved in the pathophysiology of disturbances of locomotion, gait and posture in Parkinson's disease ([Mena-Segovia et al 2004](#)). That said, the variation in nuclear connectivity between quadrupedal and bipedal animals must be considered when evaluating evidence from any non-primate animal model ([Alam et al 2011](#), [Stein and Aziz 2012](#)).)

PATHOPHYSIOLOGY OF BASAL GANGLIA DISORDERS

The basal ganglia might be said to guide intention into action. As far as their role in movement control is concerned, they appear to promote and support patterns of behaviour that are appropriate in the prevailing circumstances and to inhibit unwanted or inappropriate movements. This is exemplified by disorders of the basal ganglia, which are characterized, depending on the underlying pathology, by an inability to initiate and execute wanted movements (as in Parkinson's disease) or an inability to prevent unwanted movements (as in Huntington's disease).

Parkinson's disease is the most common pathological condition affecting the basal ganglia. It is characterized by akinesia, muscular rigidity and tremor due to degeneration of the dopaminergic neurones of the substantia nigra pars compacta (which project to the striatum in the nigrostriatal pathway). As a consequence, dopamine terminals are lost in the striatum and dopamine levels are severely depleted. Dopamine receptors, which are located on medium spiny neurones and are the target of the nigrostriatal pathway, are spared.

Dopamine appears to have a dual action on medium spiny striatal neurones. It inhibits those of the indirect pathway and excites those of the direct pathway. Consequently, when dopamine is lost from the striatum, the indirect pathway becomes overactive and the direct pathway becomes underactive ([Fig. 24.12](#)). Overactivity of the striatal projection to the external pallidum results in inhibition of pallidosubthalamic neurones and, consequently, overactivity of the subthalamic nucleus. Subthalamic efferents mediate excessive excitatory drive to the internal globus pallidus and substantia nigra pars reticularis. This is exacerbated by underactivity of the GABAergic, inhibitory direct pathway. Overactivity of basal ganglia output then inhibits the motor thalamus and its excitatory thalamocortical connections. While this description is little more than a first approximation of the underlying pathophysiology, this model of the basis of parkinsonian symptoms has led to the introduction of new neurosurgical approaches to the treatment of Parkinson's disease, based upon lesioning and deep-brain stimulation of the internal globus pallidus and subthalamic nucleus (see below).

The current medical treatment for Parkinson's disease largely relies on levodopa (L-DOPA; L-dihydroxyphenylalanine), the immediate metabolic precursor of dopamine, or on dopamine agonists. Whilst these usually provide good symptomatic relief for many years, eventually they lead to the development of side-effects, including dyskinesias. The involuntary movements that occur as a consequence of long-term

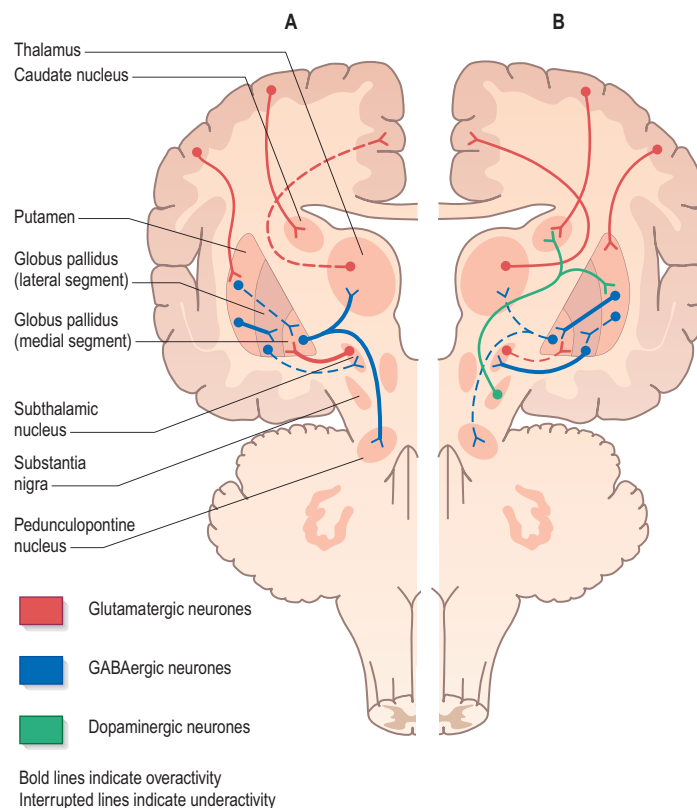


Fig. 24.12 The pathophysiology of Parkinson's disease (A) and dyskinesias (B). (Adapted with permission from Crossman AR, Neary D 2010 *Neuroanatomy*, 4th edition. Edinburgh: Churchill Livingstone.)

treatment of Parkinson's disease resemble those seen in Huntington's disease, tardive dyskinesia and ballism. Experimental evidence suggests that these may share a common neural mechanism (see [Fig. 24.12](#)). Thus, the indirect pathway becomes underactive, e.g. due to the effects of dopaminergic drugs in Parkinson's disease, or the degeneration of the striatopallidal projection to the external pallidum in Huntington's disease. This leads to physiological inhibition of the subthalamic nucleus by overactive pallidosubthalamic neurones. The involvement of the subthalamic nucleus explains why the dyskinetic movements of levodopa-induced dyskinesia and Huntington's disease resemble those of ballism produced by lesion of the subthalamic nucleus. Underactivity of the subthalamic nucleus removes the excitatory drive from internal pallidal neurones, which are known to be underactive in dyskinesias ([Crossman 1990](#)). Once again, this anatomical model of basal ganglia function is an oversimplification. Whilst it is true that underactivity of the internal globus pallidus is associated with dyskinesias, it is also known that lesions of the globus pallidus alleviate them. This so-called 'pallidotomy paradox' suggests that the dynamic aspects of pallidal and nigral efferent activity are important factors in the generation of dyskinesia ([Marsden and Obeso 1994](#)). A more recent 'rate model' of basal ganglia function hypothesizes that specific components oscillate in neuronal synchrony at different frequencies to select specific voluntary motor patterns ([Little and Brown 2014](#)). Both rodent electrophysiology and recording from deep-brain electrodes implanted in patients with Parkinson's disease suggest that neuronal activity suppression in the 8–30 Hz beta frequency band correlates with relief of parkinsonian akinesia and rigidity by both dopamine and subthalamic nucleus deep-brain stimulation.

There is evidence that dysfunction of the basal ganglia is also involved in other complex, less well understood, behavioural disorders. In animal experiments, lesions of the basal ganglia, especially of the caudate nucleus, induce uncontrollable hyperactivity (e.g. obstinate progression, incessant pacing and other constantly repeated behaviours). This and other evidence has led to the notion that the corpus striatum enables the individual to make motor choices and to avoid 'stimulus-bound' behaviour. Positron emission tomography (PET) studies in humans have shown that sufferers from obsessive-compulsive disorder (OCD), which is characterized by repeated ritualistic motor behaviour and intrusive thoughts, exhibit abnormal activity in the prefrontal cortex and caudate nuclei. There are similar suggestive observations in childhood attention deficit hyperactivity disorder (ADHD) and Gilles de la Tourette's syndrome. In this respect, it may be significant

that the basal ganglia, besides receiving connections from the frontal lobe and limbic cortices, also have an ascending influence on the prefrontal and cingulate cortices through the substantia nigra pars reticularis and dorsomedial and ventromedial thalamus in parallel, functionally segregated, corticostriothalamic circuits subserving locomotor, oculomotor, cognitive and affective behaviour (Alexander et al 1986; see Fig. 24.9).

Before the advent of levodopa therapy, neurosurgery for Parkinson's disease was commonplace. The globus pallidus and thalamus were favoured targets for chemical or thermal lesions. Pallidotomy and thalamotomy often improved rigidity and tremor, but they produced little consistent beneficial effect on akinesia. With the arrival of levodopa therapy, which very effectively alleviates akinesia, the surgical treatment of Parkinson's disease underwent a progressive decline. It soon became clear, however, that long-term use of levodopa was associated with a number of side-effects such as dyskinesias, 'wearing-off' and the 'on-off' phenomenon. More recent advances in understanding the pathophysiology of movement disorders, and of Parkinson's disease in particular, have stimulated a renaissance in the use of neurosurgery to treat movement disorders.

Lesioning the subthalamic nucleus in experimental primates that had been made parkinsonian with the neurotoxin MPTP dramatically alleviated the parkinsonian symptoms, suggesting that the subthalamic nucleus might be an appropriate clinical target (Pereira and Aziz 2006). While therapeutic surgical lesions of the subthalamic nucleus can alleviate tremor, rigidity and bradykinesia in patients with Parkinson's disease, the risk of side-effects is not trivial: the subthalamic nucleus is a small structure wrapped by fibres of passage and close to the hypothalamus and internal capsule.

In 1992, Laitinen et al reintroduced pallidotomy for the treatment of end-stage Parkinson's disease, but confined the lesions to the posteroventral part of the internal pallidum. These lesions were found to be extremely reliable in abolishing contralateral rigidity and drug-induced dyskinesias, with slightly less efficacy on tremor and bradykinesia (Laitinen et al 1992).

Implantation of deep-brain electrodes, through which high-frequency pulses generated by a pacemaker could inhibit cells in the vicinity, has been a concept since the early 1970s but did not become a widespread reality until the late 1980s, as a result of technological advances. The introduction of the technique of deep-brain stimulation (DBS), which avoids making permanent lesions, made bilateral surgery safer. There have been numerous reports of the effectiveness of both bilateral pallidal and subthalamic nucleus stimulation in Parkinson's disease (Rodriguez-Oroz et al 2012; Figs 24.13–24.14). Subthalamic nucleus stimulation is favoured by most for relieving akinesia and rigidity, with pallidal stimulation considered to ameliorate dyskinesias. Subthalamic stimulation is more effective than pallidal stimulation in allowing patients to reduce their anti-parkinsonian medication. Tremor is best relieved by stimulation of either the contralateral ventral intermediate thalamic nucleus or the zona incerta.

Another manifestation of basal ganglia dysfunction is dystonia, which is characterized by increased muscle tone and abnormal postures. Dystonia may occur as a consequence of levodopa treatment in Parkinson's disease or inherited disease (e.g. early-onset torsion, or Oppenheim's dystonia, an autosomal dominant disorder most commonly associated with a mutation in the *DYT1* gene that encodes torsin A). Although the pathophysiological basis of dystonia is unclear (Hallett 2006), it is probably caused by underactivity of basal ganglia output. The observation that painful dystonic posturing of the limbs in parkinsonian patients responds dramatically to bilateral pallidal stimulation led to the development of bilateral pallidal stimulation for dystonia. Intriguingly, in dystonia the pallidal neurones are held to fire at rates below normal, and so it is open to question how this stimulation works. Moreover, while the effect of stimulation is immediate in the case of Parkinson's disease, in dystonia the improvement may take weeks to emerge, suggesting that the neural mechanism(s) underlying the therapeutic effect of stimulation for these conditions is/are different and implicating a role for neuroplasticity in dystonia.

An even more recent neurosurgical development in the therapy of Parkinson's disease has involved targeting the pedunculopontine nucleus, which both receives fibres from and sends fibres to basal ganglia and related nuclei. Recently, DBS of the pedunculopontine nucleus through implanted electrodes has been applied in drug-resistant akinetic parkinsonian patients (Fig. 24.15). Low-frequency stimulation alleviates postural instability and on-medication gait freezing and falling, symptoms that conventional medication and surgery fail to improve.

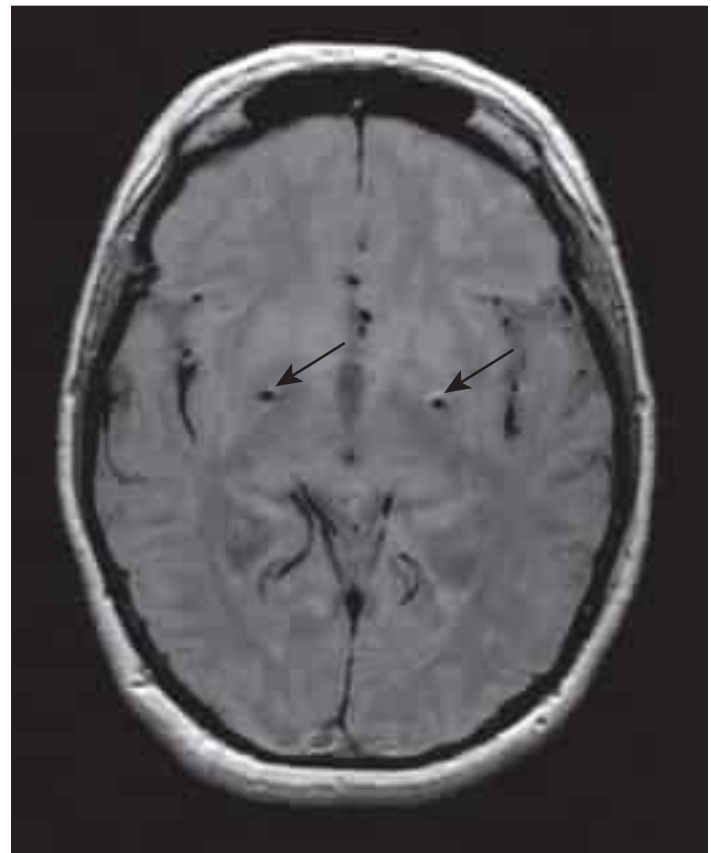


Fig. 24.13 A magnetic resonance image showing the placement of deep-brain stimulating electrodes (arrows) bilaterally in the globus pallidus of a patient with Parkinson's disease. (Courtesy of Mr L Zrinzo and Professor M Hariz, National Hospital of Neurology and Neurosurgery, London, UK.)

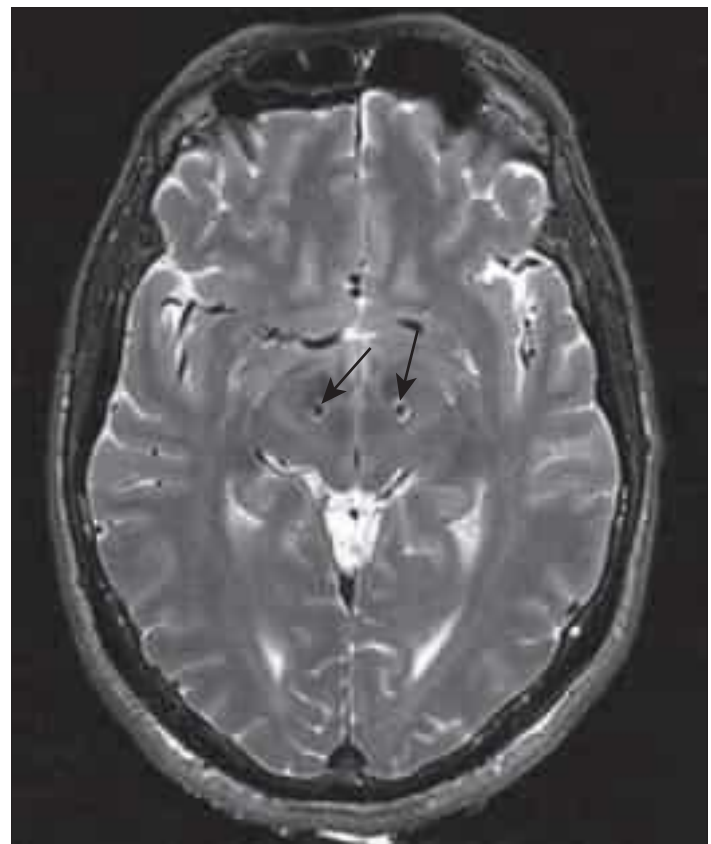


Fig. 24.14 A magnetic resonance image showing the placement of deep-brain stimulating electrodes (arrows) bilaterally in the subthalamic nucleus of a patient with Parkinson's disease. (Courtesy of Mr L Zrinzo and Professor M Hariz, National Hospital of Neurology and Neurosurgery, London, UK.)

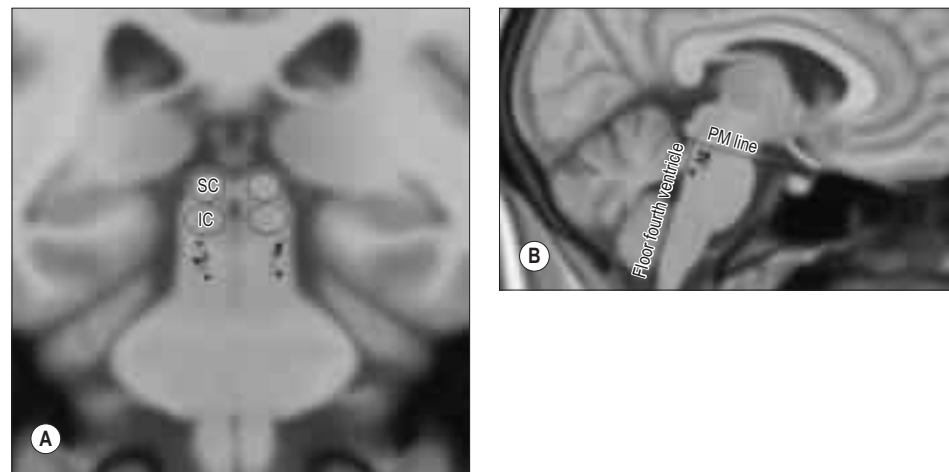


Fig. 24.15 Stimulation locations represented in Montreal Neurological Institute space in pedunculopontine nucleus deep-brain stimulation for Parkinson's disease. The relative extent of the pedunculopontine nucleus has been outlined based on choline acetyltransferase immunohistochemistry. **A**, Coronal view. **B**, Sagittal view. Abbreviations: IC, inferior colliculus; PM line, pontomesencephalic line connecting the pontomesencephalic junction to the caudal end of the inferior colliculi; SC, superior colliculus. (Courtesy of Mr J A Hyam, Departments of Neurosurgery, Oxford University Hospitals and adapted from Thevathasan et al., Alpha oscillations in the pedunculopontine nucleus correlate with gait performance in parkinsonism. *Brain*. 2012 Jan;135(Pt 1):148–60.)

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